

Basics of Computer

1. An Electronic Device That Converts Data into Information

The most important definition to understand is:

A computer is a programmable electronic device that accepts raw data as input, processes it according to a program, and produces meaningful information as output.

2. Digital Electronic Device

Your slide describes a computer as a **digital electronic device**.

Let's separate those two words.

Electronic

A computer works using **electronic circuits and electrical signals**.

Inside your computer are billions of tiny electronic components, particularly **transistors**.

These components control electrical signals to perform operations.

You don't need to think:

"The computer understands electricity."

Instead, think:

Electronic circuits manipulate signals according to instructions.

3. What Does "Digital" Mean?

This is extremely important in BSCS.

A modern computer represents information using **discrete values**, primarily:

0 and 1

This is called **binary**.

4. Two-Digit Processing — Binary

Your slide says:

Two-digit processing (Binary)

The more technically correct term is:

Binary system

"Binary" means a number system with only **two digits**:

0
1

The computer ultimately represents all kinds of information using binary patterns.

For example:

Numbers
Text
Images
Audio
Video
Programs

All can be encoded into binary data.

5. Why Does a Computer Use 0 and 1?

Because electronic circuits can reliably distinguish between different signal states.

At a simplified conceptual level:

LOW signal → 0

HIGH signal → 1

This makes it possible for electronic circuits to perform logical operations.

For example, computers use logic gates such as:

AND

OR

NOT

These are fundamental building blocks of digital computers.

You will study these much more deeply later in Computer Science.

6. A Computer Is Programmable

This is another **very important word**.

Programmable means:

A computer can be given a set of instructions that tells it what operations to perform.

Why?

Because we can give it **different programs**.

7. What Is a Program?

A **program** is a set of instructions that tells a computer what to do.

8. The Complete Definition — Understand Every Word

Now let's break your teacher's definition apart:

A computer is a programmable electronic device that accepts raw data as input and processes it with a program to produce the result as output.

"Programmable"

It can execute different sets of instructions.

"Electronic"

It operates using electronic circuits and signals.

"Device"

It is a physical machine/system designed to perform operations.

"Accepts"

It receives something from the outside world.

"Raw data"

Unprocessed facts, numbers, characters, measurements, etc.

"Input"

The data/instructions supplied to the computer.

"Processes"

The computer performs operations on the input.

"Program"

A set of instructions telling the computer what to do.

"Result"

The outcome produced after processing.

"Output"

The result delivered by the computer to the user or another system.